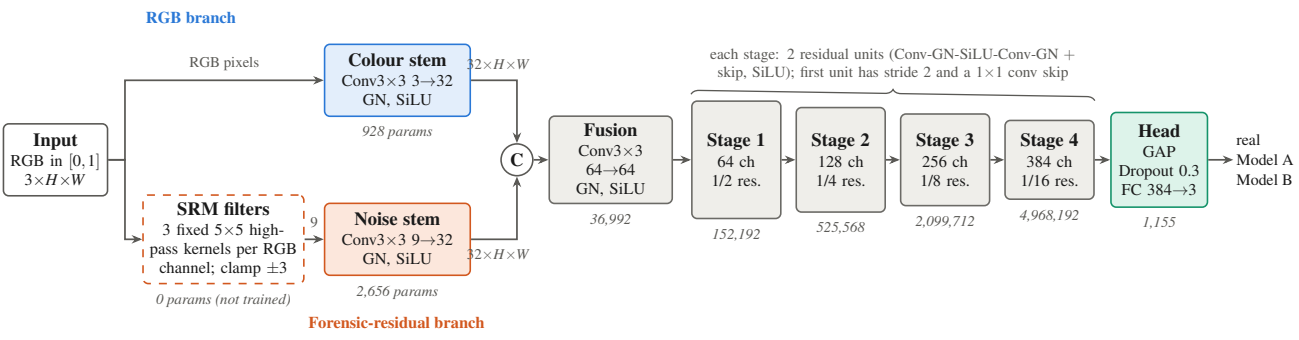

RGB branch
RGB pixels
Colour stem
Conv3×3 3→32
GN, SiLU
928 params
32×H×W
Input
RGB in [0, 1]
3×H×W
C
SRM filters
3 fixed 5×5 high-pass kernels per RGB channel; clamp ±3
0 params (not trained)
9
Noise stem
Conv3×3 9→32
GN, SiLU
2,656 params
32×H×W
Forensic-residual branch
Fusion
Conv3×3
64→64
GN, SiLU
36,992
each stage: 2 residual units (Conv-GN-SiLU-Conv-GN + skip, SiLU); first unit has stride 2 and a 1×1 conv skip
Stage 1
64 ch
1/2 res.
152,192
Stage 2
128 ch
1/4 res.
525,568
Stage 3
256 ch
1/8 res.
2,099,712
Stage 4
384 ch
1/16 res.
4,968,192
Head
GAP
Dropout 0.3
FC 384→3
1,155
real
Model A
Model B